Advanced Machine Learning, ETH Zurich

Attention

How a word reads the words around it

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Autumn semester 2026

Entry 1

Words in context



The question. In the language model, every row of E holds one word, yet the next word depends on the whole prefix. So row i must read the rows above it. How?



The question. In the language model, every row of E holds one word, yet the next word depends on the whole prefix. So row i must read the rows above it. How?

Now. Attention: every row reads the other rows and adds to itself what it finds. In Wuggish, the toy language of this lecture, every word finds what it needs in the rows above it.

Subject

A word takes its meaning from the words around it

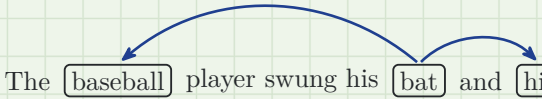
Entry 1

2

The same word can mean two different things. The words around it decide which one.

The baseball player swung his bat and hit a home run.

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Here **bat** reads baseball and hit: a club.

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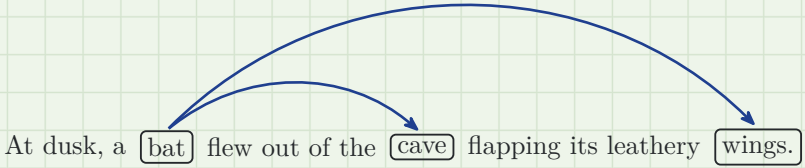
Here **bat** reads baseball and hit: a club.

At dusk, a bat flew out of the cave flapping its leathery wings.

The same word can mean two different things. The words around it decide which one.



Here **bat** reads baseball and hit: a club.



Here **bat** reads cave and wings: an animal.

Few people know the words alacrity and torpor. Read the sentence: its other words tell you what they mean.

She responded with **alacrity**, leaping to her feet with energy,

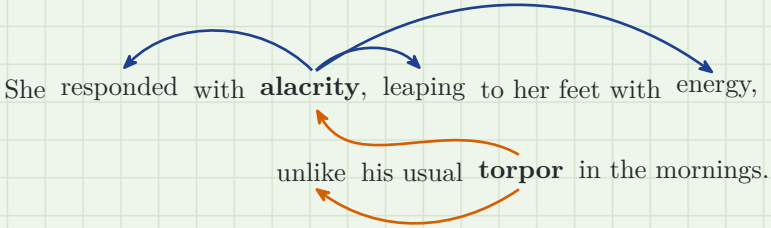
unlike his usual **torpor** in the mornings.

Few people know the words alacrity and torpor. Read the sentence: its other words tell you what they mean.



alacrity reads responded, leaping and energy: something quick and lively.

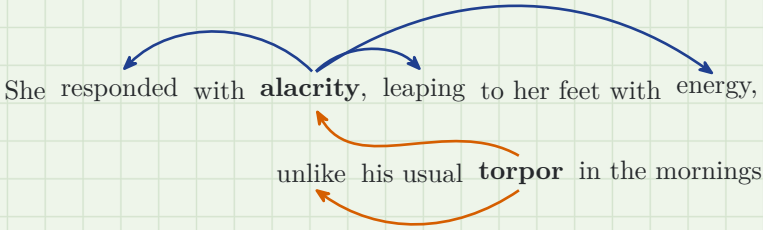
Few people know the words alacrity and torpor. Read the sentence: its other words tell you what they mean.



alacrity reads responded, leaping and energy: something quick and lively.

torpor reads unlike and alacrity: the opposite of alacrity.

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alacrity reads responded, leaping and energy: something quick and lively.

torpor reads unlike and alacrity: the opposite of alacrity.



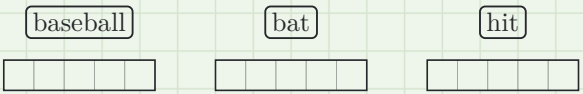
alacrity: brisk and cheerful readiness.



torpor: a state of physical or mental inactivity.

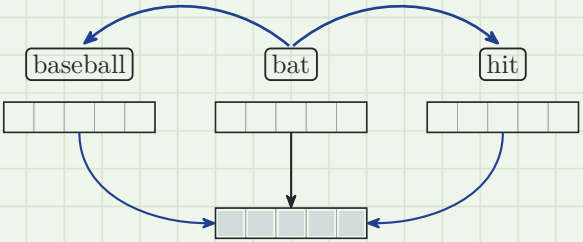
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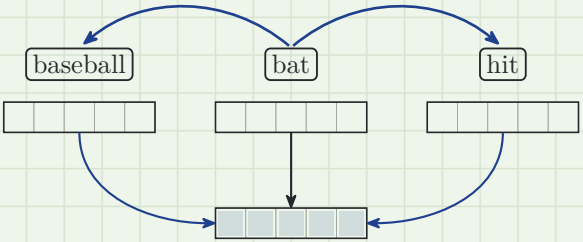


bat, now the club

2. Every word reads the other words and adds to its row what it finds.

A network that reads a sentence does it in two steps, and every word does both.

1. Every word gets a representation: a row of numbers.



bat, now the club

2. Every word reads the other words and adds to its row what it finds.

This is *attention*. The rest of the lecture computes it by hand, on a language small enough to see every number.

You go shopping for a party. What you look for is the *query*; the signs over the shops are the *keys*; what you carry home are the *values*.

Query



You go shopping for a party. What you look for is the *query*; the signs over the shops are the *keys*; what you carry home are the *values*.

Query



Keys

GreenChoice Supermarket



StoryStack Books



Threadline Clothing Co



Candy Cave



Spirit Cellar Liquors



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Values













You go shopping for a party. What you look for is the *query*; the signs over the shops are the *keys*; what you carry home are the *values*.

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Keys

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Candy Cave



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Values













The party matches three keys. Their values go home.

Entry 2

Wuggish



So far. A word takes its meaning from the words around it. Attention lets every word read the other words and add what it finds to its own row.



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Now. A toy language of eleven words, Wuggish, and a way to write each word as numbers: a row of nine features, each a 1 or a 0.

Wuggish is a made-up sister language of English. Its sentences describe a cat, a duck and a fish, each plain or with a look: running, in boots, swimming or flying.

kat





dukk







feesh





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		runnin	*bootet*		
kat					
dukk					
feesh					

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		runnin	*bootet*	*swimmin*	*flyin*
kat					
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		runnin	*bootet*	*swimmin*	*flyin*
kat			 ***pussinbootz***		
dukk		 ***runnerdukk***		 ***pondukk***	
feesh					 ***flyfeesh***

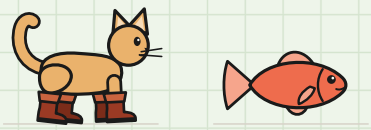
Wuggish is a made-up sister language of English. Its sentences describe a cat, a duck and a fish, each plain or with a look: running, in boots, swimming or flying.

		runnin	*bootet*	*swimmin*	*flyin*
kat			 ***pussinbootz***		
dukk		 ***runnerdukk***		 ***pondukk***	
feesh					 ***flyfeesh***

Eleven words: three things, four looks, and four things with a look. *pussinbootz* is the cat in boots that *bootet kat* draws.

Learn the eleven words from pictures, then read 12 sentences whose words come in a fresh order every time. At the end, the game shows what you have been doing all along.

bootet *feesh* *kat*



Play it now, on a laptop or a phone: scan the code, or open the address below.

eth-aml-hs26.github.io/neural-networks/viz/wuggish-game

To compute with words, we ask every word the same nine questions. Each question is a *feature*, and its answer is 1 for yes and 0 for no.

What a thing has

LEGS has legs?



FINS has fins?



WINGS has wings?



The nouns say yes here: a cat has legs, a duck legs and wings, a fish fins.



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Where a look goes

FEET goes on the feet?

FLUID goes into water or air?

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What a look brings

RUN brings a run?



BOOTS brings boots?



WATER brings water?



AIR brings air?



An adjective brings its look, and so does a word that names a thing with a look.

Ask every Wuggish word the nine questions and write its answers in a row, its *embedding*. The eleven rows form the *embedding table*.

	has			goes		brings			
	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
kat									
dukk									
feesh									
runnerdukk									
pussinbootz									
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11 words by $d = 9$ features

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	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
kat	**1**	0	0	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0	0	0
feesh	0	**1**	0	0	0	0	0	0	0
runnerdukk									
pussinbootz									
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flyfeesh									
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kat, *dukk* and *feesh* say what they have: legs; legs and wings; fins.

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pondukk	**1**	0	**1**	0	0	0	0	**1**	0
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runnin	0	0	0	**1**	0	**1**	0	0	0
bootet	0	0	0	**1**	0	0	**1**	0	0
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pondukk	**1**	0	**1**	0	0	0	0	**1**	0
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flyin	0	0	0	0	**1**	0	0	0	**1**

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kat, *dukk* and *feesh* say what they have: legs; legs and wings; fins.

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We wrote these rows by hand, to read them. A trained network learns its own rows, and its features have no names.

A word of a sentence, a *token*, is a row of the table. A feature is a column: a question that several words answer with yes.

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kat	**1**	0	0	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0	0	0
feesh	0	**1**	0	0	0	0	0	0	0
runnerdukk	**1**	0	**1**	0	0	**1**	0	0	0
pussinbootz	**1**	0	0	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	0	0	**1**	0
flyfeesh	0	**1**	0	0	0	0	0	0	**1**
runnin	0	0	0	**1**	0	**1**	0	0	0
bootet	0	0	0	**1**	0	0	**1**	0	0
swimmin	0	0	0	0	**1**	0	0	**1**	0
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pondukk	**1**	0	**1**	0	0	0	0	**1**	0
flyfeesh	0	**1**	0	0	0	0	0	0	**1**
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bootet	0	0	0	**1**	0	0	**1**	0	0
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A token: the row of *pussinbootz*, nine numbers that say a cat in boots.

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pussinbootz	**1**	0	0	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	0	0	**1**	0
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bootet	0	0	0	**1**	0	0	**1**	0	0
swimmin	0	0	0	0	**1**	0	0	**1**	0
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A token: the row of *pussinbootz*, nine numbers that say a cat in boots.

A feature: the column LEGS. Five words say yes, *kat*, *dukk*, *runnerdukk*, *pussinbootz* and *pondukk*: legs are what they share.

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bootet	0	0	0	**1**	0	0	**1**	0	0
swimmin	0	0	0	0	**1**	0	0	**1**	0
flyin	0	0	0	0	**1**	0	0	0	**1**
words	**5**	**2**	**3**	**2**	**2**	**2**	**2**	**2**	**2**

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pondukk	**1**	0	**1**	0	0	0	0	**1**	0
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runnin	0	0	0	**1**	0	**1**	0	0	0
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flyin	0	0	0	0	**1**	0	0	0	**1**
words	**5**	**2**	**3**	**2**	**2**	**2**	**2**	**2**	**2**

A token: the row of *pussinbootz*, nine numbers that say a cat in boots.

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Every column holds the 1s of two words or more: the count under it.

And there are more words than features: 11 rows, 9 columns. So a feature can never be a second name for a word.

Two rows of the table differ in one column. The row of *pussinbootz* holds the row of *kat* and one more 1, at BOOTS.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
row of *kat*	**1**	0	0	0	0	0	0	0	0





Two rows of the table differ in one column. The row of *pussinbootz* holds the row of *kat* and one more 1, at BOOTS.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
row of *kat*	**1**	0	0	0	0	0	0	0	0
+ BOOTS	0	0	0	0	0	0	**1**	0	0





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+ BOOTS	0	0	0	0	0	0	**1**	0	0
row of *pussinbootz*	**1**	0	0	0	0	0	**1**	0	0





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+ BOOTS	0	0	0	0	0	0	**1**	0	0
row of *pussinbootz*	**1**	0	0	0	0	0	**1**	0	0





The other three alike: *runnerdukk* is *dukk* plus RUN, *pondukk* is *dukk* plus WATER, and *flyfeesh* is *feesh* plus AIR.

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	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
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+ BOOTS	0	0	0	0	0	0	**1**	0	0
row of *pussinbootz*	**1**	0	0	0	0	0	**1**	0	0

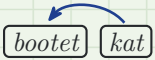




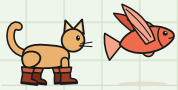
The other three alike: *runnerdukk* is *dukk* plus RUN, *pondukk* is *dukk* plus WATER, and *flyfeesh* is *feesh* plus AIR.

Keep the row of *pussinbootz* in mind: attention will arrive at it from two words.

Wuggish has one rule of order: an adjective stands somewhere before the noun it describes. Other words may stand between them, and beyond that the order carries nothing.



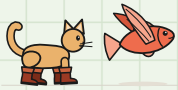
✓ a cat in boots



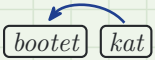
Wuggish has one rule of order: an adjective stands somewhere before the noun it describes. Other words may stand between them, and beyond that the order carries nothing.

bootet *kat* ✓ a cat in boots

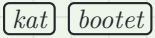
kat *bootet* ✗ the adjective stands after its noun



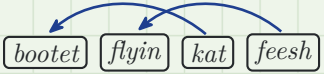
Wuggish has one rule of order: an adjective stands somewhere before the noun it describes. Other words may stand between them, and beyond that the order carries nothing.



✓ a cat in boots



✗ the adjective stands after its noun



✓ other words between them



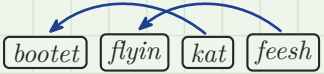
Wuggish has one rule of order: an adjective stands somewhere before the noun it describes. Other words may stand between them, and beyond that the order carries nothing.



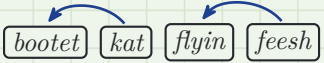
✓ a cat in boots



✗ the adjective stands after its noun



✓ other words between them



✓ the sentence we compute



Wuggish has one rule of order: an adjective stands somewhere before the noun it describes. Other words may stand between them, and beyond that the order carries nothing.



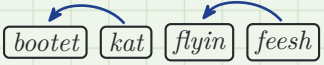
✓ a cat in boots



✗ the adjective stands after its noun



✓ other words between them



✓ the sentence we compute



So a noun finds its adjectives among the words before it: in E, in the rows above its own.

Write our sentence one word per row and replace every word by its row of the table. The result is E, with $n = 4$ rows and $d = 9$ columns.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet									
kat									
flyin									
feesh									

E: $n = 4$ rows, $d = 9$ columns

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
kat	**1**	0	0	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0	0	0
feesh	0	**1**	0	0	0	0	0	0	0
runnerdukk	**1**	0	**1**	0	0	**1**	0	0	0
pussinbootz	**1**	0	0	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	0	0	**1**	0
flyfeesh	0	**1**	0	0	0	0	0	0	**1**
runnin	0	0	0	**1**	0	**1**	0	0	0
bootet	0	0	0	**1**	0	0	**1**	0	0
swimmin	0	0	0	0	**1**	0	0	**1**	0
flyin	0	0	0	0	**1**	0	0	0	**1**

the embedding table

Write our sentence one word per row and replace every word by its row of the table. The result is E, with $n = 4$ rows and $d = 9$ columns.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet									
kat	**1**	0	0	0	0	0	0	0	0
flyin									
feesh									

E: $n = 4$ rows, $d = 9$ columns

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
kat	**1**	0	0	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0	0	0
feesh	0	**1**	0	0	0	0	0	0	0
runnerdukk	**1**	0	**1**	0	0	**1**	0	0	0
pussinbootz	**1**	0	0	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	0	0	**1**	0
flyfeesh	0	**1**	0	0	0	0	0	0	**1**
runnin	0	0	0	**1**	0	**1**	0	0	0
bootet	0	0	0	**1**	0	0	**1**	0	0
swimmin	0	0	0	0	**1**	0	0	**1**	0
flyin	0	0	0	0	**1**	0	0	0	**1**

the embedding table

Write our sentence one word per row and replace every word by its row of the table. The result is E, with $n = 4$ rows and $d = 9$ columns.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E: $n = 4$ rows, $d = 9$ columns

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
kat	**1**	0	0	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0	0	0
feesh	0	**1**	0	0	0	0	0	0	0
runnerdukk	**1**	0	**1**	0	0	**1**	0	0	0
pussinbootz	**1**	0	0	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	0	0	**1**	0
flyfeesh	0	**1**	0	0	0	0	0	0	**1**
runnin	0	0	0	**1**	0	**1**	0	0	0
bootet	0	0	0	**1**	0	0	**1**	0	0
swimmin	0	0	0	0	**1**	0	0	**1**	0
flyin	0	0	0	0	**1**	0	0	0	**1**

the embedding table

Write our sentence one word per row and replace every word by its row of the table. The result is E, with $n = 4$ rows and $d = 9$ columns.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E: $n = 4$ rows, $d = 9$ columns

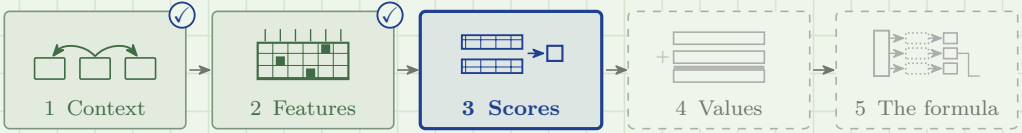
The row of *kat* holds LEGS alone: a cat. Nothing in it says booted, and *bootet* stands in the row above. Attention will make the row of *kat* read the row of *bootet*.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
kat	**1**	0	0	0	0	0	0	0	0
dukk	**1**	0	**1**	0	0	0	0	0	0
feesh	0	**1**	0	0	0	0	0	0	0
runnerdukk	**1**	0	**1**	0	0	**1**	0	0	0
pussinbootz	**1**	0	0	0	0	0	**1**	0	0
pondukk	**1**	0	**1**	0	0	0	0	**1**	0
flyfeesh	0	**1**	0	0	0	0	0	0	**1**
runnin	0	0	0	**1**	0	**1**	0	0	0
bootet	0	0	0	**1**	0	0	**1**	0	0
swimmin	0	0	0	0	**1**	0	0	**1**	0
flyin	0	0	0	0	**1**	0	0	0	**1**

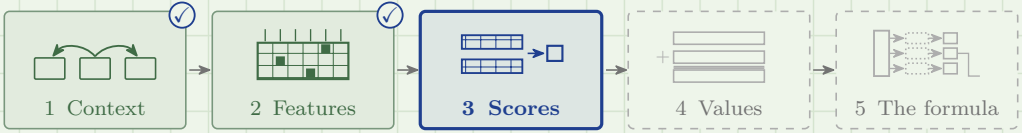
the embedding table

Entry 3

Queries, keys and scores



So far. Every word is a row of nine features, and our sentence is a block E. In E, the row of *kat* says nothing about boots.



So far. Every word is a row of nine features, and our sentence is a block E. In E, the row of *kat* says nothing about boots.

Now. Every word asks with a *query* and answers with a *key*. One number, the inner product of a query and a key, says whether two words go together.

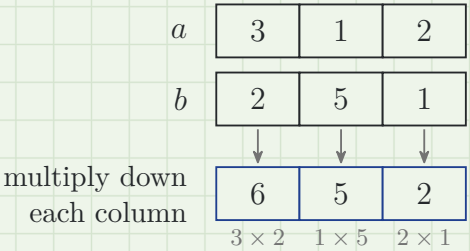
Two rows of the same length give one number, their *inner product*.

a	3	1	2

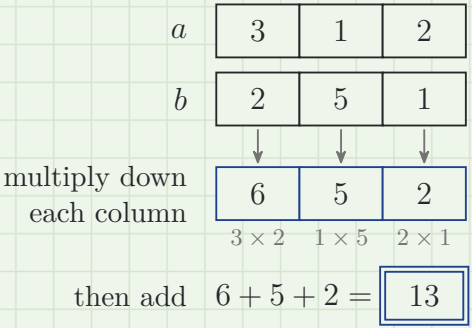
Two rows of the same length give one number, their *inner product*.

a	3	1	2
b	2	5	1

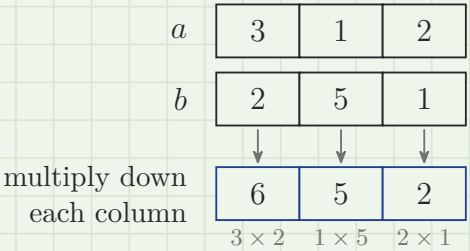
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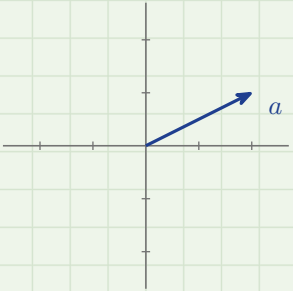


then add $6 + 5 + 2 = \boxed{13}$ the inner product of a and b

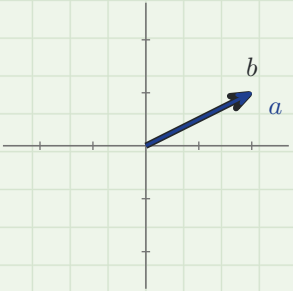
$$\text{inner product of } a \text{ and } b = a_1b_1 + a_2b_2 + a_3b_3$$

Read a row of two numbers as an arrow. Keep a and turn b: the inner product falls as the angle between them opens.

$a = (2, 1)$



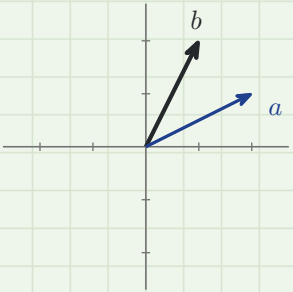
Read a row of two numbers as an arrow. Keep a and turn b: the inner product falls as the angle between them opens.



$a = (2, 1)$

$b = (2, 1)$ $\quad 2 \times 2 + 1 \times 1 = 5$ same direction

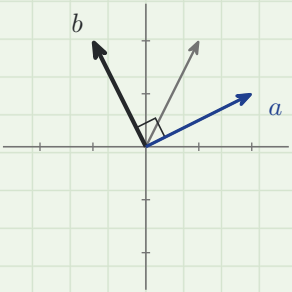
Read a row of two numbers as an arrow. Keep a and turn b: the inner product falls as the angle between them opens.



$a = (2, 1)$

$b = (2, 1)$	$2 \times 2 + 1 \times 1 = 5$	same direction
$b = (1, 2)$	$2 \times 1 + 1 \times 2 = 4$	close directions

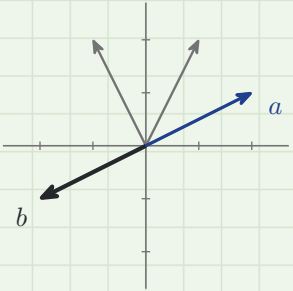
Read a row of two numbers as an arrow. Keep a and turn b: the inner product falls as the angle between them opens.



$a = (2, 1)$

$b = (2, 1)$	$2 \times 2 + 1 \times 1 = 5$	same direction
$b = (1, 2)$	$2 \times 1 + 1 \times 2 = 4$	close directions
$b = (-1, 2)$	$2 \times (-1) + 1 \times 2 = 0$	at right angles

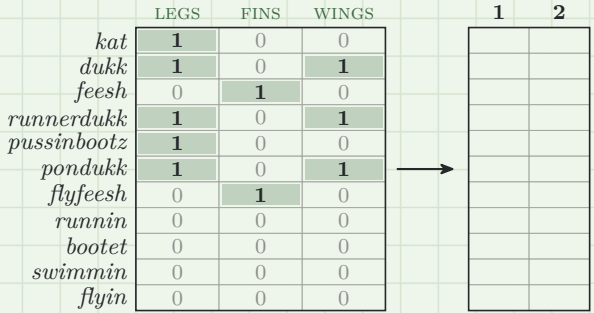
Read a row of two numbers as an arrow. Keep a and turn b: the inner product falls as the angle between them opens.



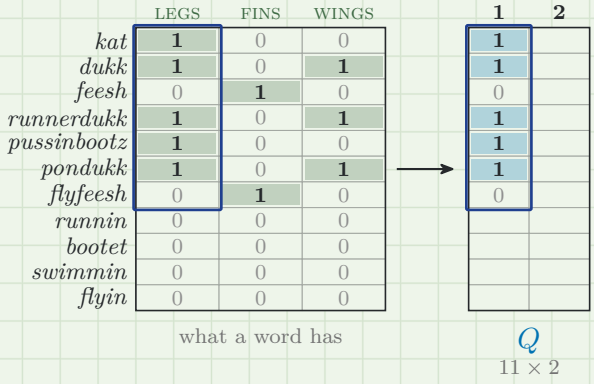
$a = (2, 1)$

$b = (2, 1)$	$2 \times 2 + 1 \times 1 = 5$	same direction
$b = (1, 2)$	$2 \times 1 + 1 \times 2 = 4$	close directions
$b = (-1, 2)$	$2 \times (-1) + 1 \times 2 = 0$	at right angles
$b = (-2, -1)$	$2 \times (-2) + 1 \times (-1) = -5$	opposite directions

Every word gets a second row, its *query*: what it looks for in the other words. A query has $k = 2$ entries of its own, slot 1 and slot 2.

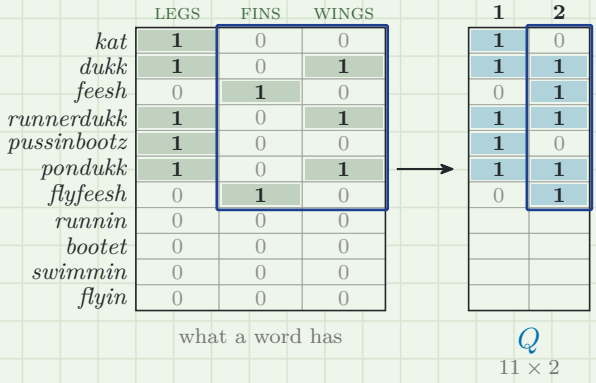


Every word gets a second row, its *query*: what it looks for in the other words. A query has $k = 2$ entries of its own, slot 1 and slot 2.



slot 1 = LEGS. A thing with legs looks for a look that goes on the feet.

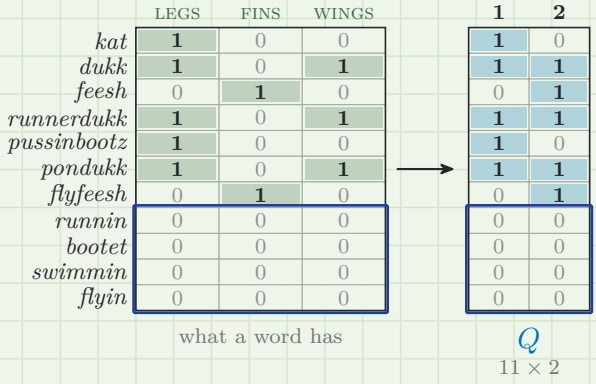
Every word gets a second row, its *query*: what it looks for in the other words. A query has $k = 2$ entries of its own, slot 1 and slot 2.



slot 1 = LEGS. A thing with legs looks for a look that goes on the feet.

slot 2 = FINS + WINGS. A thing with fins or wings looks for a look that goes into water or air.

Every word gets a second row, its *query*: what it looks for in the other words. A query has $k = 2$ entries of its own, slot 1 and slot 2.



slot 1 = LEGS. A thing with legs looks for a look that goes on the feet.

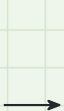
slot 2 = FINS + WINGS. A thing with fins or wings looks for a look that goes into water or air.

An adjective has none of these, so it asks for nothing: its query is $(0, 0)$.

Every word gets a second row, its *query*: what it looks for in the other words. A query has $k = 2$ entries of its own, slot 1 and slot 2.

	LEGS	FINS	WINGS
kat	1	0	0
dukk	1	0	1
feesh	0	1	0
runnerdukk	1	0	1
pussinbootz	1	0	0
pondukk	1	0	1
flyfeesh	0	1	0
runnin	0	0	0
bootet	0	0	0
swimmin	0	0	0
flyin	0	0	0

what a word has



1	2
1	0
1	1
0	1
1	1
1	0
1	1
0	1
0	0
0	0
0	0
0	0

Q

11×2

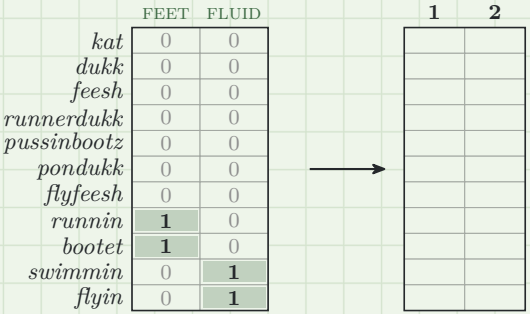
slot 1 = LEGS. A thing with legs looks for a look that goes on the feet.

slot 2 = FINS + WINGS. A thing with fins or wings looks for a look that goes into water or air.

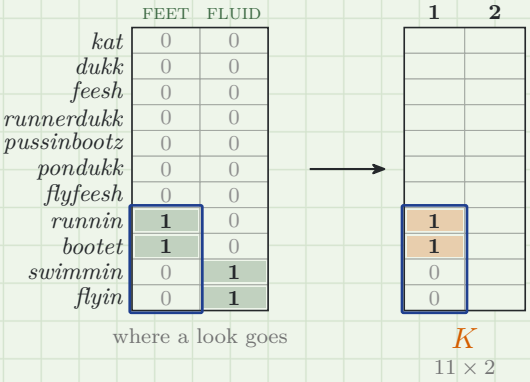
An adjective has none of these, so it asks for nothing: its query is $(0, 0)$.

The slots are numbered: they live in queries and keys, apart from the $d = 9$ features.

Every word gets a third row, its *key*: what it offers to the words that look. A key has the same two slots as a query.

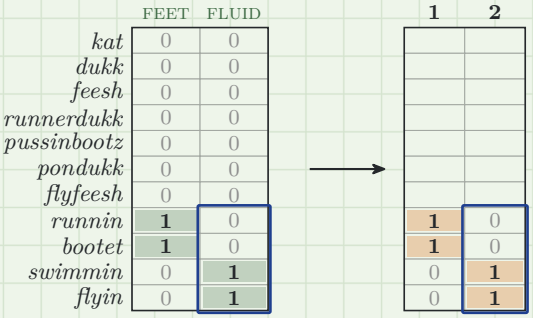


Every word gets a third row, its *key*: what it offers to the words that look. A key has the same two slots as a query.



slot 1 = FEET. *runnin* and *bootet* go on the feet, so they answer in slot 1.

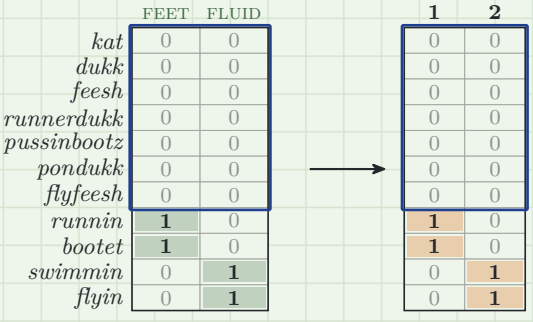
Every word gets a third row, its *key*: what it offers to the words that look. A key has the same two slots as a query.



slot 1 = FEET. *runnin* and *bootet* go on the feet, so they answer in slot 1.

slot 2 = FLUID. *swimmin* and *flyin* go into water or air, so they answer in slot 2.

Every word gets a third row, its *key*: what it offers to the words that look. A key has the same two slots as a query.



slot 1 = FEET. *runnin* and *bootet* go on the feet, so they answer in slot 1.

slot 2 = FLUID. *swimmin* and *flyin* go into water or air, so they answer in slot 2.

A noun says no to both questions: its key is $(0, 0)$, also for *runnerdukk*, *pussinbootz*, *pondukk* and *flyfeesh*.

Every word gets a third row, its *key*: what it offers to the words that look. A key has the same two slots as a query.

	FEET	FLUID
kat	0	0
dukk	0	0
feesh	0	0
runnerdukk	0	0
pussinbootz	0	0
pondukk	0	0
flyfeesh	0	0
runnin	**1**	0
bootet	**1**	0
swimmin	0	**1**
flyin	0	**1**

where a look goes

1	**2**
0	0
0	0
0	0
0	0
0	0
0	0
0	0
1	0
1	0
0	**1**
0	**1**

K

11×2

slot 1 = FEET. *runnin* and *bootet* go on the feet, so they answer in slot 1.

slot 2 = FLUID. *swimmin* and *flyin* go into water or air, so they answer in slot 2.

A noun says no to both questions: its key is $(0, 0)$, also for *runnerdukk*, *pussinbootz*, *pondukk* and *flyfeesh*.

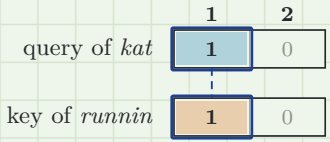
So slot 1 pairs things with legs and looks for the feet, slot 2 things with fins or wings and looks for water or air.

Take the query of *kat* and the key of an adjective. Multiply slot by slot and add: every slot where both rows hold a 1 adds 1.

	1	2
query of *kat*	1	0

inner product

Take the query of *kat* and the key of an adjective. Multiply slot by slot and add: every slot where both rows hold a 1 adds 1.



inner product

1

$1 \times 1 + 0 \times 0 = 1$: both fill slot 1

Take the query of *kat* and the key of an adjective. Multiply slot by slot and add: every slot where both rows hold a 1 adds 1.

	1	2	inner product	
query of *kat*	1	0		
key of *runnin*	1	0	1	$1 \times 1 + 0 \times 0 = 1$: both fill slot 1
key of *bootet*	1	0	1	the same

Take the query of *kat* and the key of an adjective. Multiply slot by slot and add: every slot where both rows hold a 1 adds 1.

	1	2	inner product	
query of *kat*	1	0		
key of *runnin*	1	0	1	$1 \times 1 + 0 \times 0 = 1$: both fill slot 1
key of *bootet*	1	0	1	the same
key of *swimmin*	0	1	0	$1 \times 0 + 0 \times 1 = 0$: no slot in common

Take the query of *kat* and the key of an adjective. Multiply slot by slot and add: every slot where both rows hold a 1 adds 1.

	1	2	inner product	
query of *kat*	1	0		
key of *runnin*	1	0	1	$1 \times 1 + 0 \times 0 = 1$: both fill slot 1
key of *bootet*	1	0	1	the same
key of *swimmin*	0	1	0	$1 \times 0 + 0 \times 1 = 0$: no slot in common
key of *flyin*	0	1	0	the same

Take the query of *kat* and the key of an adjective. Multiply slot by slot and add: every slot where both rows hold a 1 adds 1.

	1	2	inner product	
query of *kat*	1	0		
key of *runnin*	1	0	1	$1 \times 1 + 0 \times 0 = 1$: both fill slot 1
key of *bootet*	1	0	1	the same
key of *swimmin*	0	1	0	$1 \times 0 + 0 \times 1 = 0$: no slot in common
key of *flyin*	0	1	0	the same

So a cat runs or wears boots and never swims or flies. The same sums let a fish swim or fly, and a duck do all four: the grammar of Wuggish.

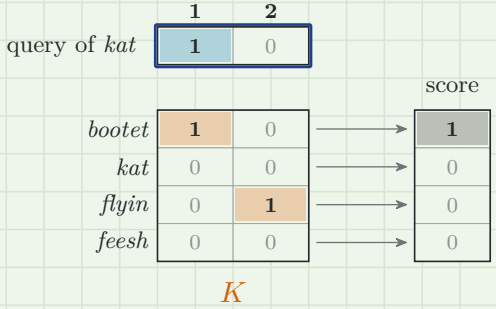
Lay the query of *kat* over every key of our sentence and take the inner products. Each gives a *score*: how well that word fits *kat*.

	1	2
query of *kat*	**1**	0

bootet	**1**	0
kat	0	0
flyin	0	**1**
feesh	0	0

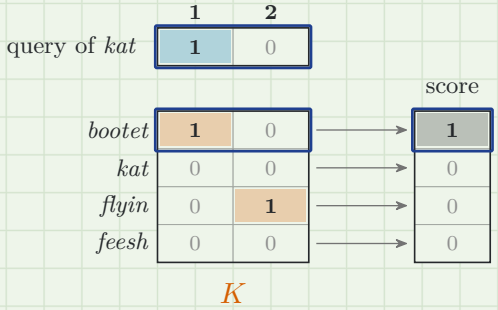
K

Lay the query of *kat* over every key of our sentence and take the inner products. Each gives a *score*: how well that word fits *kat*.



score of *kat* with word j = query of *kat* · key of word j

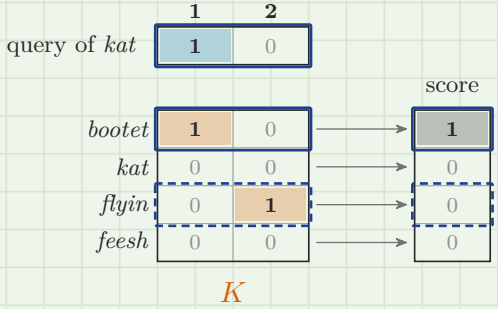
Lay the query of *kat* over every key of our sentence and take the inner products. Each gives a *score*: how well that word fits *kat*.



bootet, the word above *kat*, goes on the feet, and a cat has legs: score 1.

score of *kat* with word j = query of *kat* $\cdot$ key of word j

Lay the query of *kat* over every key of our sentence and take the inner products. Each gives a *score*: how well that word fits *kat*.

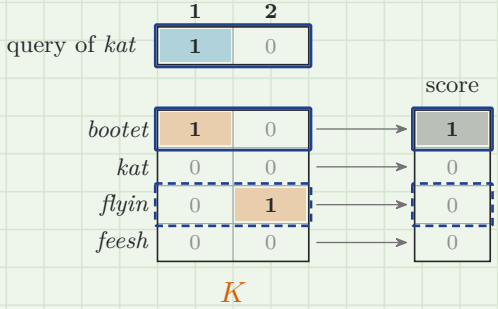


bootet, the word above *kat*, goes on the feet, and a cat has legs: score 1.

flyin, the word after *kat*, goes into the air, and a cat has no fins or wings: score 0.

score of *kat* with word j = query of *kat* · key of word j

Lay the query of *kat* over every key of our sentence and take the inner products. Each gives a *score*: how well that word fits *kat*.



bootet, the word above *kat*, goes on the feet, and a cat has legs: score 1.

flyin, the word after *kat*, goes into the air, and a cat has no fins or wings: score 0.

kat and *feesh* are nouns, and their keys are all zeros: score 0.

score of *kat* with word j = query of *kat* · key of word j

Stack the queries into Q and lay the keys on their side, $K^\top$. Their product holds every score at once: row i, column j is the query of word i against the key of word j.

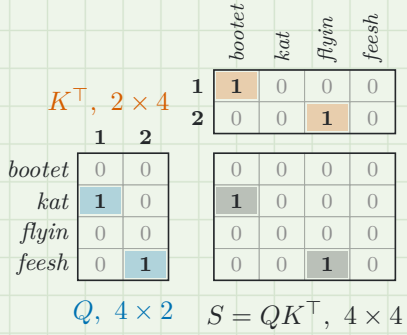
$K^\top$, 2×4

	1	2
bootet	0	0
kat	**1**	0
flyin	0	0
feesh	0	**1**

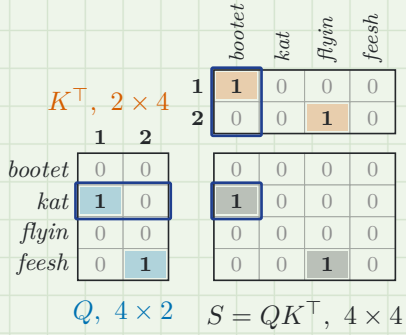
Q, 4×2

	bootet	*kat*	*flyin*	*feesh*
1	**1**	0	0	0
2	0	0	**1**	0

Stack the queries into Q and lay the keys on their side, $K^\top$. Their product holds every score at once: row i, column j is the query of word i against the key of word j.

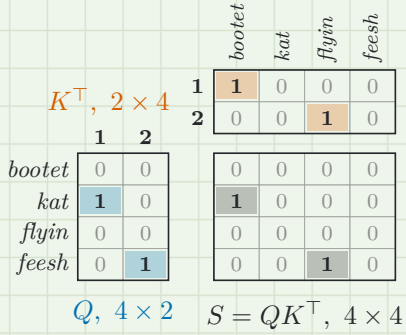


Stack the queries into Q and lay the keys on their side, $K^\top$. Their product holds every score at once: row i, column j is the query of word i against the key of word j.



Row *kat*, column *bootet*: the query of *kat* against the key of *bootet*, score 1.

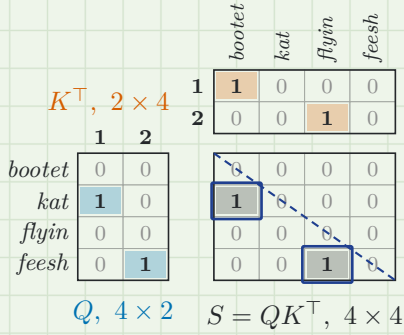
Stack the queries into Q and lay the keys on their side, $K^\top$. Their product holds every score at once: row i, column j is the query of word i against the key of word j.



Row *kat*, column *bootet*: the query of *kat* against the key of *bootet*, score 1.

Two 1s, one for each noun. An adjective asks for nothing, so its row is 0.

Stack the queries into Q and lay the keys on their side, $K^\top$. Their product holds every score at once: row i, column j is the query of word i against the key of word j.

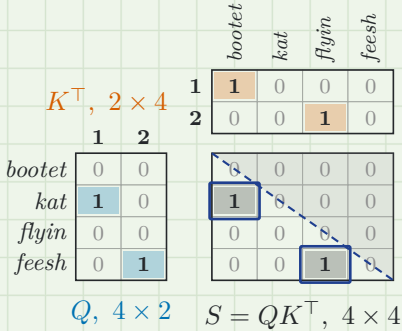


Row *kat*, column *bootet*: the query of *kat* against the key of *bootet*, score 1.

Two 1s, one for each noun. An adjective asks for nothing, so its row is 0.

Both 1s lie below the diagonal: a noun reads a word above it, since an adjective stands before its noun.

Stack the queries into Q and lay the keys on their side, $K^\top$. Their product holds every score at once: row i, column j is the query of word i against the key of word j.



Row *kat*, column *bootet*: the query of *kat* against the key of *bootet*, score 1.

Two 1s, one for each noun. An adjective asks for nothing, so its row is 0.

Both 1s lie below the diagonal: a noun reads a word above it, since an adjective stands before its noun.

Attention as computed here reads every row. A language model masks the shaded scores, where a row would read a row below it: in Wuggish they are all 0, so the mask changes nothing here. It comes with the transformer.

Entry 4

Values



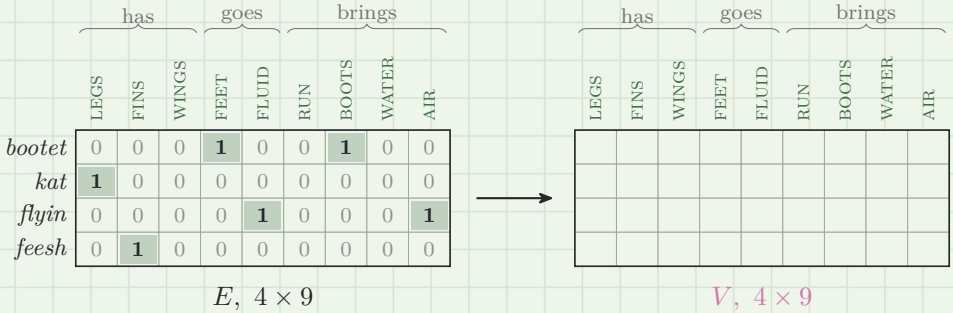
So far. The scores pair *kat* with *bootet* and *feesh* with *flyin*. Each noun has found its adjective, in a row above its own.



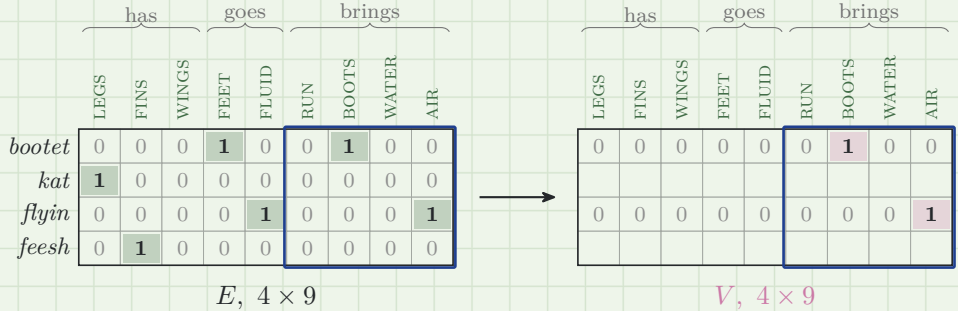
So far. The scores pair *kat* with *bootet* and *feesh* with *flyin*. Each noun has found its adjective, in a row above its own.

Now. Every word gives a *value*, what its look brings, and every word adds to its own row the values of the words it scored 1 with.

Every word gets a fourth row, its *value*: what it adds to a word that reads it. A value keeps the four features of what a look brings and sets the other five to 0.



Every word gets a fourth row, its *value*: what it adds to a word that reads it. A value keeps the four features of what a look brings and sets the other five to 0.



E, 4×9 V, 4×9

bootet gives BOOTS and *flyin* gives AIR: the brings part of their rows. Where a look goes stays out of the value; the key used it.

Every word gets a fourth row, its *value*: what it adds to a word that reads it. A value keeps the four features of what a look brings and sets the other five to 0.

	has			goes		brings				
	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E, 4×9

	has			goes		brings				
	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
	0	0	0	0	0	0	**1**	0	0
	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	**1**
	0	0	0	0	0	0	0	0	0

V, 4×9

bootet gives BOOTS and *flyin* gives AIR: the brings part of their rows. Where a look goes stays out of the value; the key used it.

A plain noun brings no look: the values of *kat* and *feesh* are all zeros.

kat adds to its row the value of every word of the sentence, each times its score. Only *bootet* scored 1.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
row of *kat* in E	**1**	0	0	0	0	0	0	0	0

kat adds to its row the value of every word of the sentence, each times its score. Only *bootet* scored 1.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
row of *kat* in E	**1**	0	0	0	0	0	0	0	0
$+ 1 \times$ value of *bootet*	0	0	0	0	0	0	**1**	0	0

kat adds to its row the value of every word of the sentence, each times its score. Only *bootet* scored 1.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
row of *kat* in E	**1**	0	0	0	0	0	0	0	0
$+ 1 \times$ value of *bootet*	0	0	0	0	0	0	**1**	0	0
$+ 0 \times$ every other value	0	0	0	0	0	0	0	0	0

kat adds to its row the value of every word of the sentence, each times its score. Only *bootet* scored 1.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
row of *kat* in E	**1**	0	0	0	0	0	0	0	0
$+ 1 \times$ value of *bootet*	0	0	0	0	0	0	**1**	0	0
$+ 0 \times$ every other value	0	0	0	0	0	0	0	0	0
new row of *kat*	**1**	0	0	0	0	0	**1**	0	0

LEGS and BOOTS: a cat in boots. Every number is still a 0 or a 1.

kat adds to its row the value of every word of the sentence, each times its score. Only *bootet* scored 1.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
row of *kat* in E	**1**	0	0	0	0	0	0	0	0
$+ 1 \times$ value of *bootet*	0	0	0	0	0	0	**1**	0	0
$+ 0 \times$ every other value	0	0	0	0	0	0	0	0	0
new row of *kat*	**1**	0	0	0	0	0	**1**	0	0

LEGS and BOOTS: a cat in boots. Every number is still a 0 or a 1.

new row of *kat* = row of *kat* $+ \sum_j$ score of *kat* with word $j \times$ value of word j

Set the new row of *kat* under the row that the table gives *pussinbootz*, and compare them column by column.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
new row of *kat* after attention	**1**	0	0	0	0	0	**1**	0	0





Set the new row of *kat* under the row that the table gives *pussinbootz*, and compare them column by column.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
new row of *kat* after attention	**1**	0	0	0	0	0	**1**	0	0
row of *pussinbootz* from the table	**1**	0	0	0	0	0	**1**	0	0





Set the new row of *kat* under the row that the table gives *pussinbootz*, and compare them column by column.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
new row of *kat* after attention	**1**	0	0	0	0	0	**1**	0	0
	=	=	=	=	=	=	=	=	=
row of *pussinbootz* from the table	**1**	0	0	0	0	0	**1**	0	0

the same





Set the new row of *kat* under the row that the table gives *pussinbootz*, and compare them column by column.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
new row of *kat* after attention	1	0	0	0	0	0	1	0	0
	=	=	=	=	=	=	=	=	=
row of *pussinbootz* from the table	1	0	0	0	0	0	1	0	0

the same





The same nine numbers, and the same picture: a cat in boots.

Set the new row of *kat* under the row that the table gives *pussinbootz*, and compare them column by column.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
new row of *kat* after attention	1	0	0	0	0	0	1	0	0
	=	=	=	=	=	=	=	=	=
row of *pussinbootz* from the table	1	0	0	0	0	0	1	0	0

the same





The same nine numbers, and the same picture: a cat in boots.

A word is a row, and a phrase can arrive at the same row. After one layer of attention, *bootet kat* reads as *pussinbootz*.

The other noun does the same. *feesh* scored 1 with *flyin*, the word above it, and adds its value, AIR.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
row of *feesh* in E	0	**1**	0	0	0	0	0	0	0





The other noun does the same. *feesh* scored 1 with *flyin*, the word above it, and adds its value, AIR.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
row of *feesh* in E	0	**1**	0	0	0	0	0	0	0
$+ 1 \times$ value of *flyin*	0	0	0	0	0	0	0	0	**1**





The other noun does the same. *feesh* scored 1 with *flyin*, the word above it, and adds its value, AIR.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
row of *feesh* in E	0	**1**	0	0	0	0	0	0	0
$+ 1 \times$ value of *flyin*	0	0	0	0	0	0	0	0	**1**
new row of *feesh*	0	**1**	0	0	0	0	0	0	**1**





The other noun does the same. *feesh* scored 1 with *flyin*, the word above it, and adds its value, AIR.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
row of *feesh* in E	0	**1**	0	0	0	0	0	0	0
$+\ 1 \times$ value of *flyin*	0	0	0	0	0	0	0	0	**1**
new row of *feesh*	0	**1**	0	0	0	0	0	0	**1**
	=	=	=	=	=	=	=	=	=
row of *flyfeesh*	0	**1**	0	0	0	0	0	0	**1**





The other noun does the same. *feesh* scored 1 with *flyin*, the word above it, and adds its value, AIR.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
row of *feesh* in E	0	**1**	0	0	0	0	0	0	0
$+\ 1 \times$ value of *flyin*	0	0	0	0	0	0	0	0	**1**
new row of *feesh*	0	**1**	0	0	0	0	0	0	**1**
	=	=	=	=	=	=	=	=	=
row of *flyfeesh*	0	**1**	0	0	0	0	0	0	**1**





FINS and AIR: after attention, *flyin feesh* reads as *flyfeesh*, the flying fish.

Our sentence on one sheet: its queries, keys and values side by side. Then one word at a time: its scores, and the values it adds to its own row.

bootet
kat
flyin
feesh

Our sentence on one sheet: its queries, keys and values side by side. Then one word at a time: its scores, and the values it adds to its own row.

Queries

	1	2
bootet	0	0
kat	**1**	0
flyin	0	0
feesh	0	**1**

Q

Our sentence on one sheet: its queries, keys and values side by side. Then one word at a time: its scores, and the values it adds to its own row.

Queries **Keys**

	1	2
bootet	0	0
kat	**1**	0
flyin	0	0
feesh	0	**1**

Q

	1	2
bootet	**1**	0
kat	0	0
flyin	0	**1**
feesh	0	0

K

Our sentence on one sheet: its queries, keys and values side by side. Then one word at a time: its scores, and the values it adds to its own row.

Queries

	1	2
bootet	0	0
kat	**1**	0
flyin	0	0
feesh	0	**1**

Q

Keys

	1	2
bootet	**1**	0
kat	0	0
flyin	0	**1**
feesh	0	0

K

Values

	has			goes		brings			
	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	0	0	0	**1**	0	0
kat	0	0	0	0	0	0	0	0	0
flyin	0	0	0	0	0	0	0	0	**1**
feesh	0	0	0	0	0	0	0	0	0

V

Our sentence on one sheet: its queries, keys and values side by side. Then one word at a time: its scores, and the values it adds to its own row.

Queries

	1	2
bootet	0	0
kat	1	0
flyin	0	0
feesh	0	1

Q

Keys

	1	2
bootet	1	0
kat	0	0
flyin	0	1
feesh	0	0

K

Values

	has			goes		brings			
	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	0	0	0	1	0	0
kat	0	0	0	0	0	0	0	0	0
flyin	0	0	0	0	0	0	0	0	1
feesh	0	0	0	0	0	0	0	0	0

V

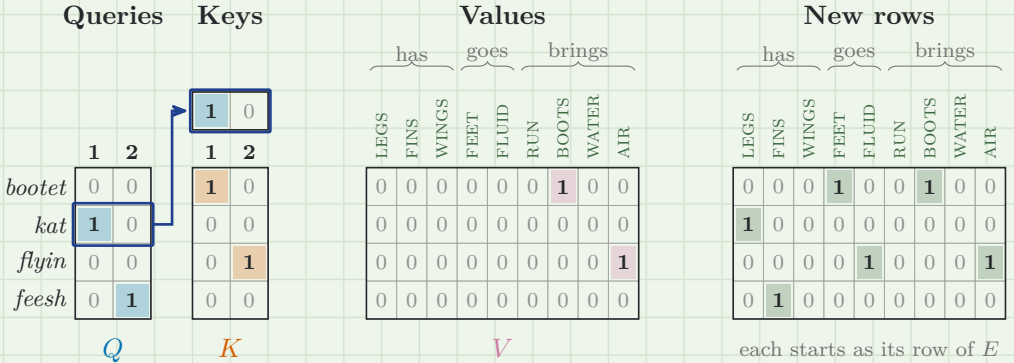
New rows

	has			goes		brings			
	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	1	0	0	1	0	0
kat	1	0	0	0	0	0	0	0	0
flyin	0	0	0	0	1	0	0	0	1
feesh	0	1	0	0	0	0	0	0	0

each starts as its row of E

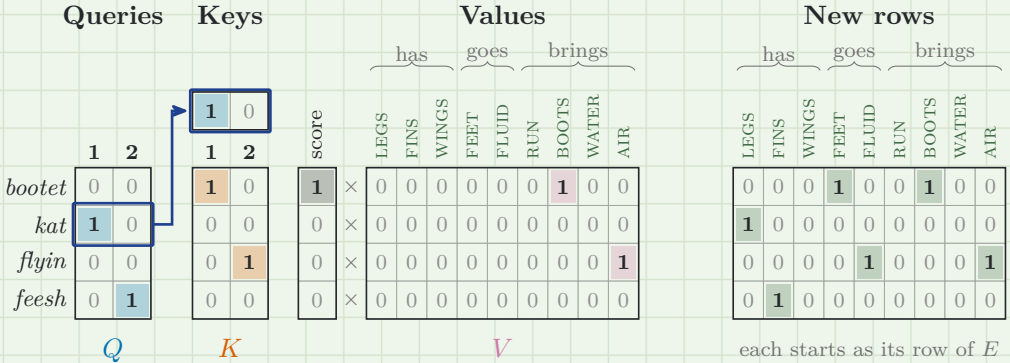
The new rows start as the rows of E: every word keeps its own row.

Our sentence on one sheet: its queries, keys and values side by side. Then one word at a time: its scores, and the values it adds to its own row.



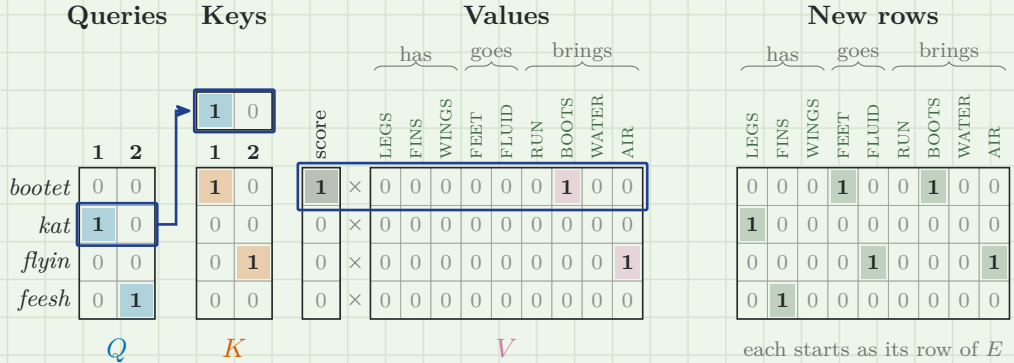
kat asks with its query: lay it over every key and multiply slot by slot.

Our sentence on one sheet: its queries, keys and values side by side. Then one word at a time: its scores, and the values it adds to its own row.



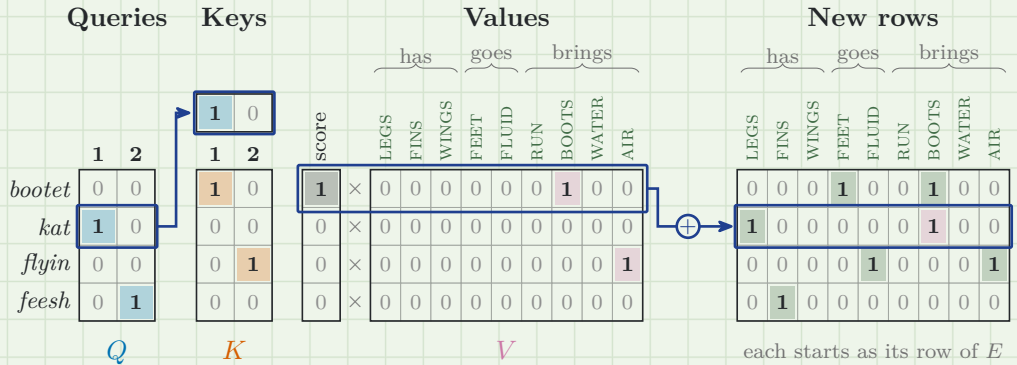
The scores of *kat*: 1 with *bootet*, the word above it, and 0 with every other word.

Our sentence on one sheet: its queries, keys and values side by side. Then one word at a time: its scores, and the values it adds to its own row.



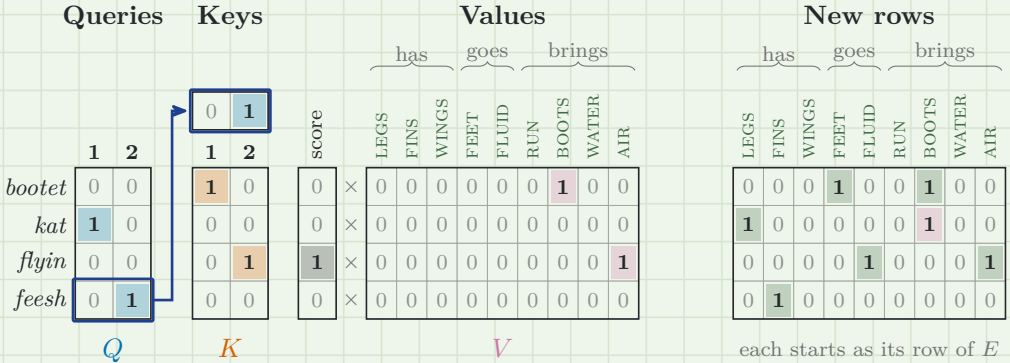
Each score multiplies the value of its word, so only the value of *bootet* counts: BOOTS.

Our sentence on one sheet: its queries, keys and values side by side. Then one word at a time: its scores, and the values it adds to its own row.



kat adds BOOTS to its own row: LEGS and BOOTS, the row of *pussinbootz*.

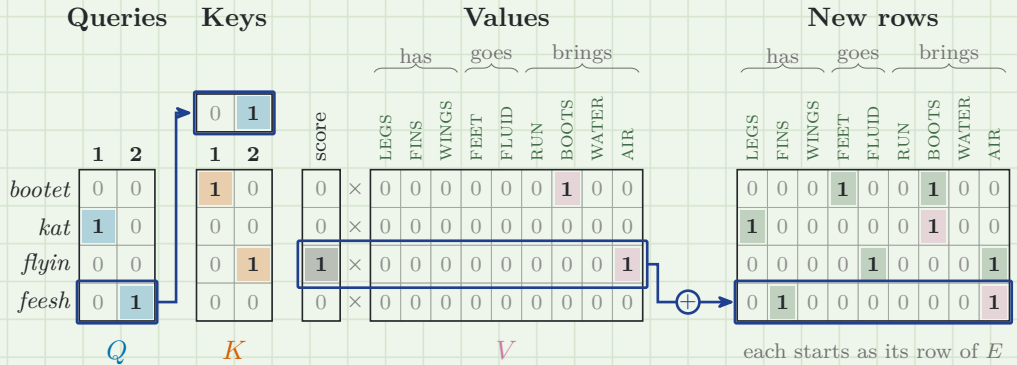
Our sentence on one sheet: its queries, keys and values side by side. Then one word at a time: its scores, and the values it adds to its own row.



kat adds BOOTS to its own row: LEGS and BOOTS, the row of *pussinbootz*.

feesh asks in slot 2 and scores 1 with *flyin* alone, the word above it.

Our sentence on one sheet: its queries, keys and values side by side. Then one word at a time: its scores, and the values it adds to its own row.



kat adds BOOTS to its own row: LEGS and BOOTS, the row of *pussinbootz*.
feesh adds AIR, the value of *flyin*: FINS and AIR, the row of *flyfeesh*.

Our sentence on one sheet: its queries, keys and values side by side. Then one word at a time: its scores, and the values it adds to its own row.

Queries

	1	2
bootet	0	0
kat	**1**	0
flyin	0	0
feesh	0	**1**

Q

Keys

	1	2
bootet	**1**	0
kat	0	0
flyin	0	**1**
feesh	0	0

K

Values

	has			goes		brings			
	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	0	0	0	**1**	0	0
kat	0	0	0	0	0	0	0	0	0
flyin	0	0	0	0	0	0	0	0	**1**
feesh	0	0	0	0	0	0	0	0	0

V

New rows

	has			goes		brings			
	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	**1**	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	**1**

each starts as its row of E

kat adds BOOTS to its own row: LEGS and BOOTS, the row of *pussinbootz*.

feesh adds AIR, the value of *flyin*: FINS and AIR, the row of *flyfeesh*.

An adjective asks for nothing: its query is $(0, 0)$, every score is 0, and its row stays as it was.

Row i of SV adds up the rows of V, each times its score in row i of S. One product does it for every word at once.

	bootet	*kat*	*flyin*	*feesh*
bootet	0	0	0	0
kat	**1**	0	0	0
flyin	0	0	0	0
feesh	0	0	**1**	0

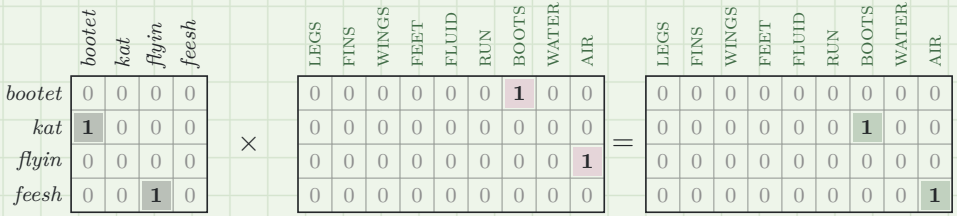
S, 4×4

$\times$

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	0	0	0	**1**	0	0
kat	0	0	0	0	0	0	0	0	0
flyin	0	0	0	0	0	0	0	0	**1**
feesh	0	0	0	0	0	0	0	0	0

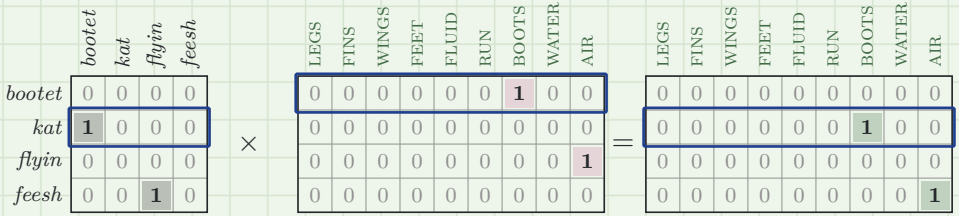
V, 4×9

Row i of SV adds up the rows of V, each times its score in row i of S. One product does it for every word at once.



$S,\ 4 \times 4$ $V,\ 4 \times 9$ $SV,\ 4 \times 9$

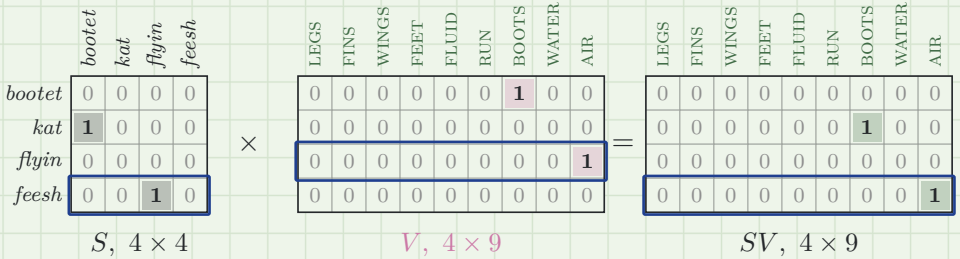
Row i of SV adds up the rows of V, each times its score in row i of S. One product does it for every word at once.



S, 4×4 V, 4×9 SV, 4×9

Row *kat* of S holds a 1 at *bootet*, so row *kat* of SV is the value of *bootet*: BOOTS.

Row i of SV adds up the rows of V, each times its score in row i of S. One product does it for every word at once.



S, 4×4 V, 4×9 SV, 4×9

Row *kat* of S holds a 1 at *bootet*, so row *kat* of SV is the value of *bootet*: BOOTS.

Row *feesh* of S holds a 1 at *flyin*, so row *feesh* of SV is AIR.

Add SV to E, row by row. A noun keeps its own features and gains the look of its adjective. An adjective found nothing, so it keeps its row as it was.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E

$+$

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	**1**	0	0
	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	**1**

SV

Add SV to E, row by row. A noun keeps its own features and gains the look of its adjective. An adjective found nothing, so it keeps its row as it was.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E

+

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	**1**	0	0
	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	**1**

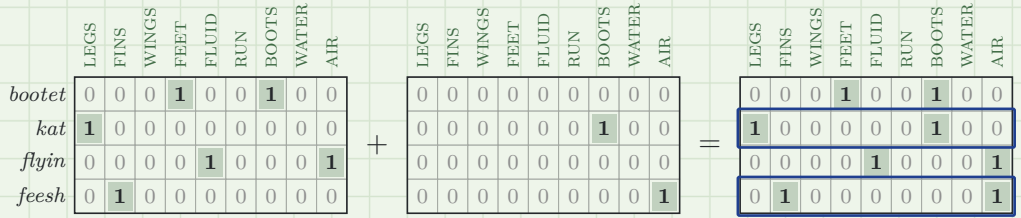
SV

=

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
	0	0	0	**1**	0	0	**1**	0	0
	1	0	0	0	0	0	**1**	0	0
	0	0	0	0	**1**	0	0	0	**1**
	0	**1**	0	0	0	0	0	0	**1**

$E + SV$, 4×9

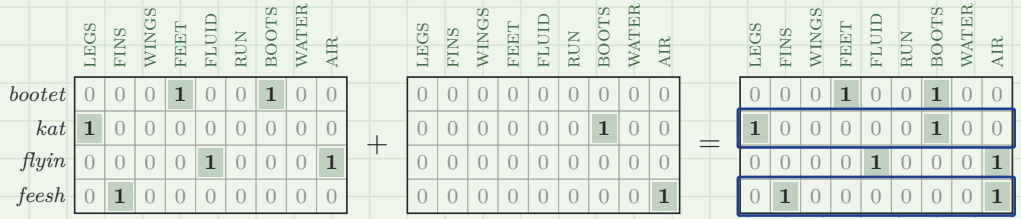
Add SV to E, row by row. A noun keeps its own features and gains the look of its adjective. An adjective found nothing, so it keeps its row as it was.



E SV $E + SV$, 4×9

The two nouns now carry the looks of their adjectives, and every number is still 0 or 1.

Add SV to E, row by row. A noun keeps its own features and gains the look of its adjective. An adjective found nothing, so it keeps its row as it was.



E SV $E + SV$, 4×9

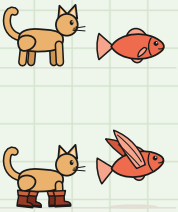
The two nouns now carry the looks of their adjectives, and every number is still 0 or 1.

Adding E back, so that every word keeps its own row, is the *residual connection*.

Read the new rows of *kat* and *feesh* against the table. One layer of attention has read the sentence: each adjective has found its noun.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	**1**	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	**1**

$E + SV$



Read the new rows of *kat* and *feesh* against the table. One layer of attention has read the sentence: each adjective has found its noun.

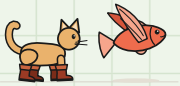
	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	**1**	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	**1**

the row of *pussinbootz*

the row of *flyfeesh*

$E + SV$





Read the new rows of *kat* and *feesh* against the table. One layer of attention has read the sentence: each adjective has found its noun.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR	
bootet	0	0	0	**1**	0	0	**1**	0	0	unchanged
kat	**1**	0	0	0	0	0	**1**	0	0	the row of *pussinbootz*
flyin	0	0	0	0	**1**	0	0	0	**1**	unchanged
feesh	0	**1**	0	0	0	0	0	0	**1**	the row of *flyfeesh*

$E + SV$





Read the new rows of *kat* and *feesh* against the table. One layer of attention has read the sentence: each adjective has found its noun.

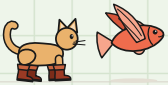
	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR	
bootet	0	0	0	**1**	0	0	**1**	0	0	unchanged
kat	**1**	0	0	0	0	0	**1**	0	0	the row of *pussinbootz*
flyin	0	0	0	0	**1**	0	0	0	**1**	unchanged
feesh	0	**1**	0	0	0	0	0	0	**1**	the row of *flyfeesh*

$E + SV$

before: the rows of E



after: the rows of $E + SV$



Read the new rows of *kat* and *feesh* against the table. One layer of attention has read the sentence: each adjective has found its noun.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR	
bootet	0	0	0	**1**	0	0	**1**	0	0	unchanged
kat	**1**	0	0	0	0	0	**1**	0	0	the row of *pussinbootz*
flyin	0	0	0	0	**1**	0	0	0	**1**	unchanged
feesh	0	**1**	0	0	0	0	0	0	**1**	the row of *flyfeesh*

$E + SV$

before: the rows of E



after: the rows of $E + SV$

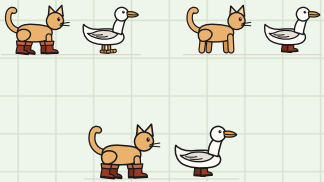


One layer of attention: n rows in, n rows out, each row enriched by the rows it scored 1 with. *bootet kat flyin feesh* now reads as *pussinbootz* and *flyfeesh*.

In *bootet kat dukk* one of the two things wears boots: Puss in Boots and a duck, or a cat and a duck in boots. Compute the scores and the new rows.

	bootet	*kat*	*dukk*
bootet	0	0	0
kat	**1**	0	0
dukk	**1**	0	0

$S,\ 3 \times 3$

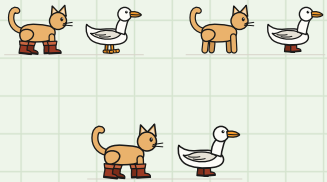


In *bootet kat dukk* one of the two things wears boots: Puss in Boots and a duck, or a cat and a duck in boots. Compute the scores and the new rows.

	bootet	*kat*	*dukk*
bootet	0	0	0
kat	**1**	0	0
dukk	**1**	0	0

$S,\ 3 \times 3$

Both nouns have legs, so both score 1 with *bootet*: a tie.

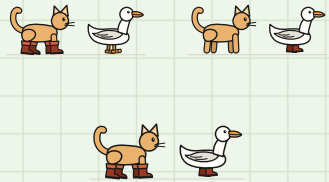


In *bootet kat dukk* one of the two things wears boots: Puss in Boots and a duck, or a cat and a duck in boots. Compute the scores and the new rows.

	bootet	*kat*	*dukk*
bootet	0	0	0
kat	**1**	0	0
dukk	**1**	0	0

$S,\ 3 \times 3$

Both nouns have legs, so both score 1 with *bootet*: a tie.



	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR	
kat	**1**	0	0	0	0	0	**1**	0	0	the row of *pussinbootz*
dukk	**1**	0	**1**	0	0	0	**1**	0	0	no word

new rows of $E + SV$: both in boots

In *bootet kat dukk* one of the two things wears boots: Puss in Boots and a duck, or a cat and a duck in boots. Compute the scores and the new rows.

	bootet	*kat*	*dukk*
bootet	0	0	0
kat	**1**	0	0
dukk	**1**	0	0

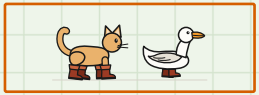
$S,\ 3 \times 3$

Both nouns have legs, so both score 1 with *bootet*: a tie.

what a reader means, one of



what attention draws



	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR	
kat	**1**	0	0	0	0	0	**1**	0	0	the row of *pussinbootz*
dukk	**1**	0	**1**	0	0	0	**1**	0	0	no word

new rows of $E + SV$: both in boots

In *bootet kat dukk* one of the two things wears boots: Puss in Boots and a duck, or a cat and a duck in boots. Compute the scores and the new rows.

	bootet	*kat*	*dukk*
bootet	0	0	0
kat	**1**	0	0
dukk	**1**	0	0

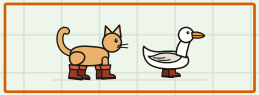
$S,\ 3 \times 3$

Both nouns have legs, so both score 1 with *bootet*: a tie.

what a reader means, one of



what attention draws



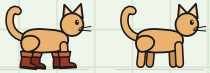
	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR	
kat	**1**	0	0	0	0	0	**1**	0	0	the row of *pussinbootz*
dukk	**1**	0	**1**	0	0	0	**1**	0	0	no word

new rows of $E + SV$: both in boots

Nothing in the words settles it, and attention sees no order at all. The transformer adds the position of every word to its row.

Could the keys be read off the looks themselves, a run and boots in slot 1, water and air in slot 2? Try *pussinbootz kat*, Puss in Boots and a cat.



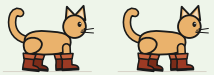


Could the keys be read off the looks themselves, a run and boots in slot 1, water and air in slot 2? Try *pussinbootz kat*, Puss in Boots and a cat.

Keys read off the looks

	1	2
key of *pussinbootz*	1	0

It is the key of *bootet*. The query of *kat* scores 1 with it, and *kat* takes the value of *pussinbootz*, BOOTS:



Puss in Boots and a cat in boots

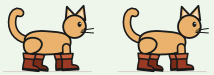


Could the keys be read off the looks themselves, a run and boots in slot 1, water and air in slot 2? Try *pussinbootz kat*, Puss in Boots and a cat.

Keys read off the looks

	1	2
key of *pussinbootz*	1	0

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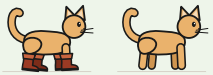


Puss in Boots and a cat in boots

Keys from where a look goes

	1	2
key of *pussinbootz*	0	0

It is all zeros: a noun goes neither on the feet nor into water or air. *kat* reads nothing, and both rows stay as they were:



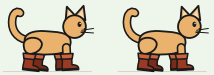
Puss in Boots and a cat

Could the keys be read off the looks themselves, a run and boots in slot 1, water and air in slot 2? Try *pussinbootz kat*, Puss in Boots and a cat.

Keys read off the looks

	1	2
key of *pussinbootz*	1	0

It is the key of *bootet*. The query of *kat* scores 1 with it, and *kat* takes the value of *pussinbootz*, BOOTS:

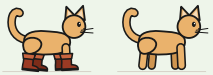


Puss in Boots and a cat in boots

Keys from where a look goes

	1	2
key of *pussinbootz*	0	0

It is all zeros: a noun goes neither on the feet nor into water or air. *kat* reads nothing, and both rows stay as they were:



Puss in Boots and a cat

pussinbootz carries boots and must keep them: only the adjectives say where a look goes.

Entry 5

Attention in one formula



So far. We wrote every query, key and value by hand, and every new row came out as a 0 or a 1.



So far. We wrote every query, key and value by hand, and every new row came out as a 0 or a 1.

Now. A network computes Q, K and V from E with three blocks of weights, and turns every row of scores into weights that add up to 1.

A network does not get the queries for free: it computes them from E, with a block of weights W^Q. Here every weight is a 0 or a 1, so we can read them.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E, 4×9

A network does not get the queries for free: it computes them from E, with a block of weights W^Q. Here every weight is a 0 or a 1, so we can read them.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E, 4×9

$\times$

	1	2
LEGS	**1**	0
FINS	0	**1**
WINGS	0	**1**
FEET	0	0
FLUID	0	0
RUN	0	0
BOOTS	0	0
WATER	0	0
AIR	0	0

W^Q, 9×2

A network does not get the queries for free: it computes them from E, with a block of weights W^Q. Here every weight is a 0 or a 1, so we can read them.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E, 4×9

$\times$

	1	2
LEGS	**1**	0
FINS	0	**1**
WINGS	0	**1**
FEET	0	0
FLUID	0	0
RUN	0	0
BOOTS	0	0
WATER	0	0
AIR	0	0

W^Q, 9×2

$=$

	1	2
	0	0
	1	0
	0	0
	0	**1**

Q, 4×2

A network does not get the queries for free: it computes them from E, with a block of weights W^Q. Here every weight is a 0 or a 1, so we can read them.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

$E,\ 4 \times 9$

$\times$

	1	2
LEGS	**1**	0
FINS	0	**1**
WINGS	0	**1**
FEET	0	0
FLUID	0	0
RUN	0	0
BOOTS	0	0
WATER	0	0
AIR	0	0

$W^Q,\ 9 \times 2$

$=$

	1	2
	0	0
	1	0
	0	0
	0	**1**

$Q,\ 4 \times 2$

Row LEGS of W^Q holds a 1 in slot 1, rows FINS and WINGS a 1 in slot 2: the rules of the queries, as a block of weights. The row of *kat* has legs, so it asks in slot 1.

The keys come from E the same way, with a second block of weights, W^K. Its rows say in which slot each feature answers.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E, 4×9

$\times$

	1	**2**
LEGS	0	0
FINS	0	0
WINGS	0	0
FEET	**1**	0
FLUID	0	**1**
RUN	0	0
BOOTS	0	0
WATER	0	0
AIR	0	0

W^K, 9×2

The keys come from E the same way, with a second block of weights, W^K. Its rows say in which slot each feature answers.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

$E,\ 4 \times 9$

$\times$

	1	2
LEGS	0	0
FINS	0	0
WINGS	0	0
FEET	**1**	0
FLUID	0	**1**
RUN	0	0
BOOTS	0	0
WATER	0	0
AIR	0	0

$W^K,\ 9 \times 2$

$=$

1	2
1	0
0	0
0	**1**
0	0

$K,\ 4 \times 2$

The keys come from E the same way, with a second block of weights, W^K. Its rows say in which slot each feature answers.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E, 4×9

$\times$

	1	2
LEGS	0	0
FINS	0	0
WINGS	0	0
FEET	**1**	0
FLUID	0	**1**
RUN	0	0
BOOTS	0	0
WATER	0	0
AIR	0	0

W^K, 9×2

$=$

	1	2
bootet	**1**	0
kat	0	0
flyin	0	**1**
feesh	0	0

K, 4×2

Row FEET, slot 1: a look that goes on the feet answers where things with legs ask. Row FLUID, slot 2: where things with fins or wings ask. So a cat runs or wears boots, a fish swims or flies, and a duck does all four.

The keys come from E the same way, with a second block of weights, W^K. Its rows say in which slot each feature answers.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E, 4×9

$\times$

	1	2
LEGS	0	0
FINS	0	0
WINGS	0	0
FEET	**1**	0
FLUID	0	**1**
RUN	0	0
BOOTS	0	0
WATER	0	0
AIR	0	0

W^K, 9×2

$=$

	1	2
	1	0
	0	0
	0	**1**
	0	0

K, 4×2

Training must find such weights in the text.

Row FEET, slot 1: a look that goes on the feet answers where things with legs ask. Row FLUID, slot 2: where things with fins or wings ask. So a cat runs or wears boots, a fish swims or flies, and a duck does all four.

A third block of weights, W^V, makes the values. It keeps the four features of what a look brings and drops the other five.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E, 4×9

$\times$

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
LEGS	0	0	0	0	0	0	0	0	0
FINS	0	0	0	0	0	0	0	0	0
WINGS	0	0	0	0	0	0	0	0	0
FEET	0	0	0	0	0	0	0	0	0
FLUID	0	0	0	0	0	0	0	0	0
RUN	0	0	0	0	0	**1**	0	0	0
BOOTS	0	0	0	0	0	0	**1**	0	0
WATER	0	0	0	0	0	0	0	**1**	0
AIR	0	0	0	0	0	0	0	0	**1**

W^V, 9×9

A third block of weights, W^V, makes the values. It keeps the four features of what a look brings and drops the other five.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E, 4×9

$\times$

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
LEGS	0	0	0	0	0	0	0	0	0
FINS	0	0	0	0	0	0	0	0	0
WINGS	0	0	0	0	0	0	0	0	0
FEET	0	0	0	0	0	0	0	0	0
FLUID	0	0	0	0	0	0	0	0	0
RUN	0	0	0	0	0	**1**	0	0	0
BOOTS	0	0	0	0	0	0	**1**	0	0
WATER	0	0	0	0	0	0	0	**1**	0
AIR	0	0	0	0	0	0	0	0	**1**

W^V, 9×9

$=$

LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
0	0	0	0	0	0	**1**	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	**1**
0	0	0	0	0	0	0	0	0

V, 4×9

A third block of weights, W^V, makes the values. It keeps the four features of what a look brings and drops the other five.

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
bootet	0	0	0	**1**	0	0	**1**	0	0
kat	**1**	0	0	0	0	0	0	0	0
flyin	0	0	0	0	**1**	0	0	0	**1**
feesh	0	**1**	0	0	0	0	0	0	0

E, 4×9

$\times$

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
LEGS	0	0	0	0	0	0	0	0	0
FINS	0	0	0	0	0	0	0	0	0
WINGS	0	0	0	0	0	0	0	0	0
FEET	0	0	0	0	0	0	0	0	0
FLUID	0	0	0	0	0	0	0	0	0
RUN	0	0	0	0	0	**1**	0	0	0
BOOTS	0	0	0	0	0	0	**1**	0	0
WATER	0	0	0	0	0	0	0	**1**	0
AIR	0	0	0	0	0	0	0	0	**1**

W^V, 9×9

$=$

LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
0	0	0	0	0	0	**1**	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	**1**
0	0	0	0	0	0	0	0	0

V, 4×9

A 1 on the diagonal at RUN, BOOTS, WATER and AIR, and 0 everywhere else: the value of *bootet* is BOOTS, the value of a plain noun all zeros. These are the values we used.

A real attention layer divides every row of scores by its total. In our sentence each row that asks reads one word, so nothing changes. Here is a row that reads two.

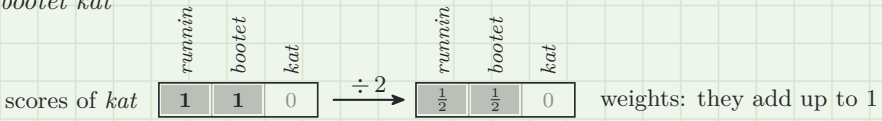
runnin bootet kat

	runnin	*bootet*	*kat*
scores of *kat*	**1**	**1**	0



A real attention layer divides every row of scores by its total. In our sentence each row that asks reads one word, so nothing changes. Here is a row that reads two.

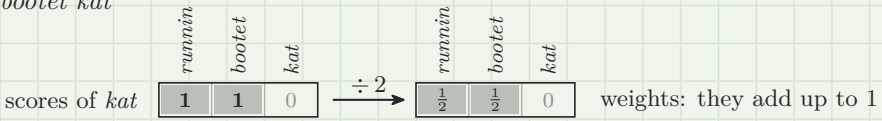
runnin bootet kat

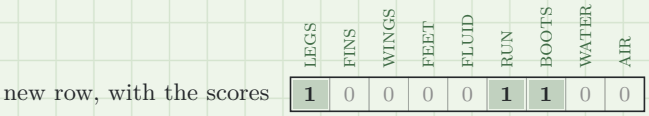




A real attention layer divides every row of scores by its total. In our sentence each row that asks reads one word, so nothing changes. Here is a row that reads two.

runnin bootet kat

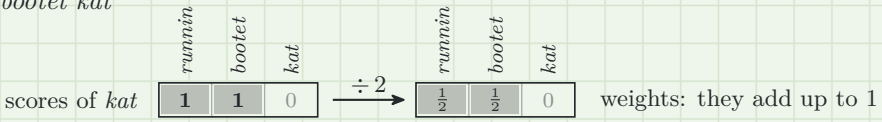


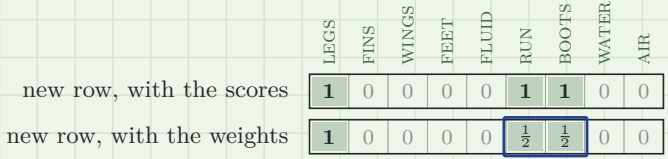




A real attention layer divides every row of scores by its total. In our sentence each row that asks reads one word, so nothing changes. Here is a row that reads two.

runnin bootet kat







A real attention layer divides every row of scores by its total. In our sentence each row that asks reads one word, so nothing changes. Here is a row that reads two.

runnin bootet kat

	runnin	*bootet*	*kat*		*runnin*	*bootet*	*kat*	
scores of *kat*	**1**	**1**	0	$\div 2$	$\frac{1}{2}$	$\frac{1}{2}$	0	weights: they add up to 1

	LEGS	FINS	WINGS	FEET	FLUID	RUN	BOOTS	WATER	AIR
new row, with the scores	**1**	0	0	0	0	**1**	**1**	0	0
new row, with the weights	**1**	0	0	0	0	$\frac{1}{2}$	$\frac{1}{2}$	0	0



With weights the cat takes half of each look: the same features, at half strength. We kept the raw scores so that every number stayed a 0 or a 1. A row of zeros stays 0.

Instead of dividing the scores themselves, a transformer divides e^{score} by its total, the *softmax*. Every weight is then positive, and a large score takes almost everything.

$$\text{weight of word } j = \frac{e^{s_j}}{\sum_l e^{s_l}}$$

bootet *kat* *flyin* *feesh*

Instead of dividing the scores themselves, a transformer divides e^{score} by its total, the *softmax*. Every weight is then positive, and a large score takes almost everything.

$$\text{weight of word } j = \frac{e^{s_j}}{\sum_l e^{s_l}}$$

	bootet	*kat*	*flyin*	*feesh*
scores of *kat*	**0.40**	0.20	0.20	0.20

With our scores of 0 and 1 the softmax barely chooses.

Instead of dividing the scores themselves, a transformer divides e^{score} by its total, the *softmax*. Every weight is then positive, and a large score takes almost everything.

$$\text{weight of word } j = \frac{e^{s_j}}{\sum_l e^{s_l}}$$

	bootet	*kat*	*flyin*	*feesh*
scores of *kat*	**0.40**	0.20	0.20	0.20
with a score of 9	**0.99**	0.00	0.00	0.00

With our scores of 0 and 1 the softmax barely chooses.

With weights three times larger, a match scores 9 and takes almost all.

Instead of dividing the scores themselves, a transformer divides e^{score} by its total, the *softmax*. Every weight is then positive, and a large score takes almost everything.

$$\text{weight of word } j = \frac{e^{s_j}}{\sum_l e^{s_l}}$$

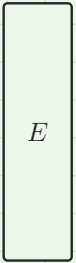
	bootet	*kat*	*flyin*	*feesh*
scores of *kat*	**0.40**	0.20	0.20	0.20
with a score of 9	**0.99**	0.00	0.00	0.00

With our scores of 0 and 1 the softmax barely chooses.

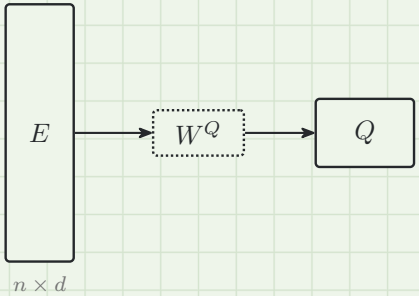
With weights three times larger, a match scores 9 and takes almost all.

A trained network learns large scores for the words that belong together. Every score is first divided by $\sqrt{k}$, so that longer queries do not make it larger. A row of zeros, an adjective's, gets 0.25 for each word: an even mix of the values, which nothing reads in this layer.

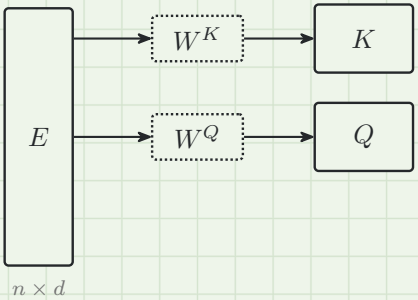
Three blocks of weights make Q, K and V from E. The softmax turns the scores into weights, the weights mix the values into A, and A is added to E.



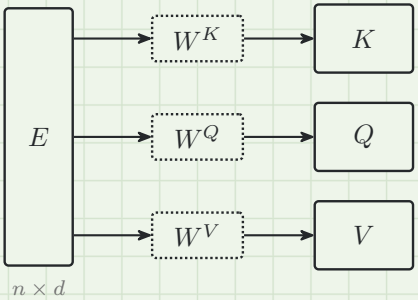
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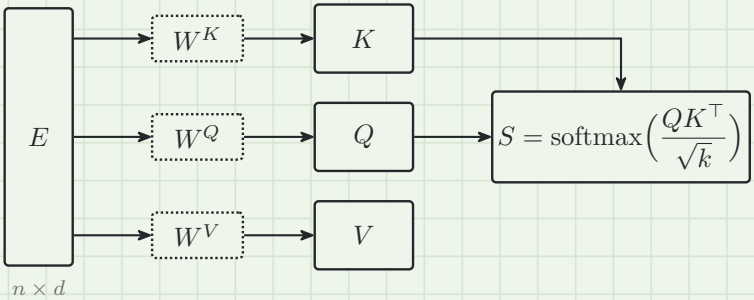
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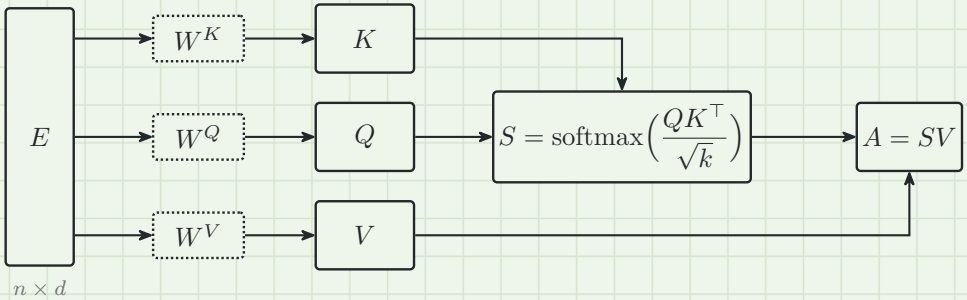
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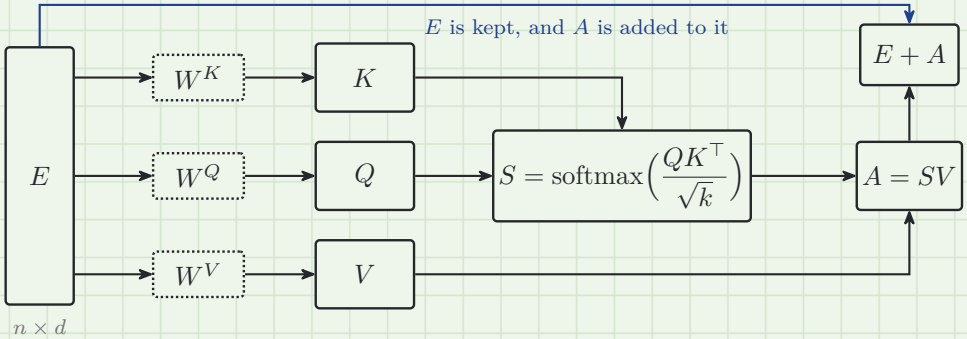
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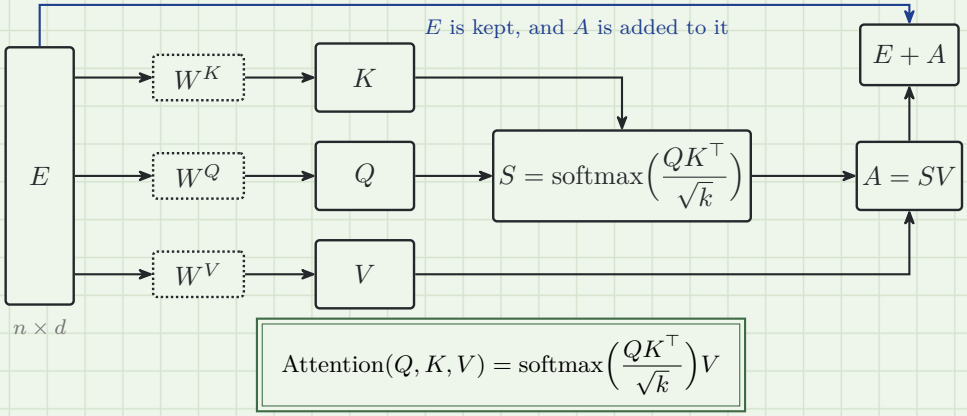
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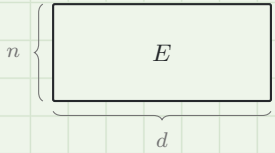
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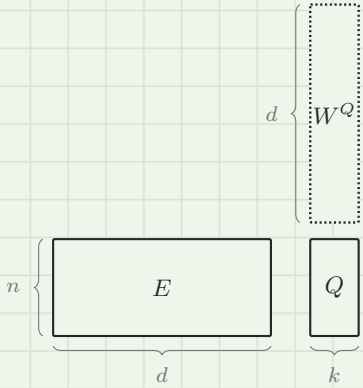
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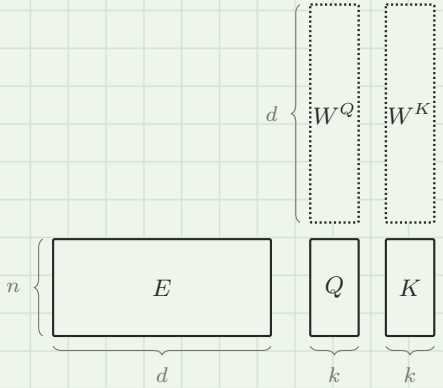
Every block drawn to scale, with the sizes of our sentence: $n = 4$ words, $d = 9$ features, and $k = 2$ slots in a query and a key.



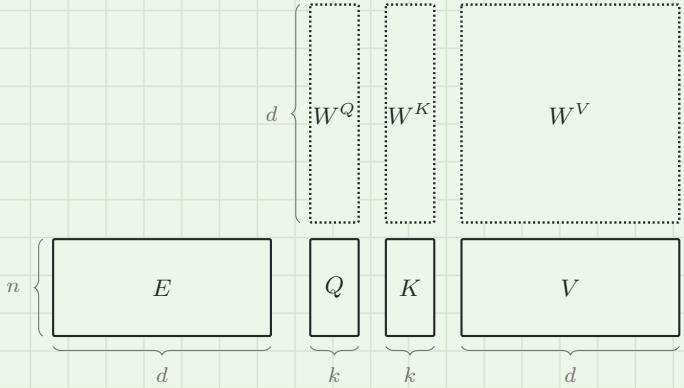
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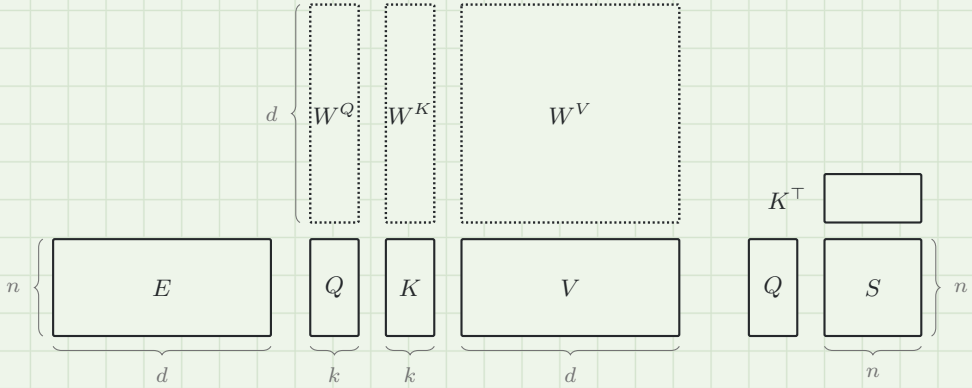
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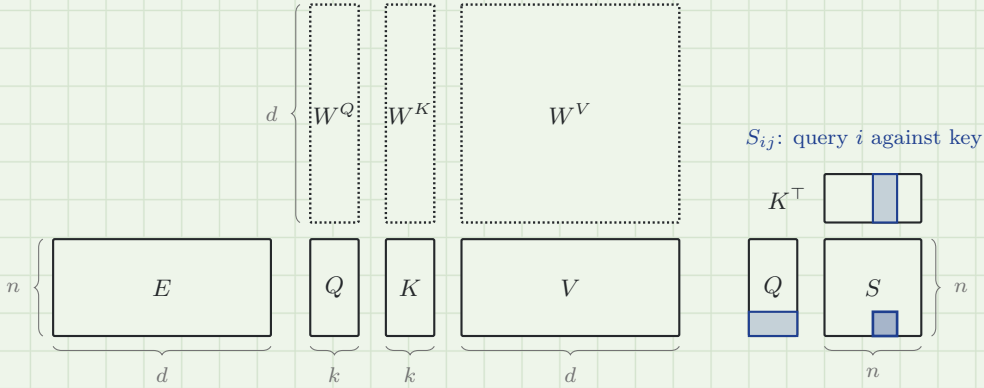
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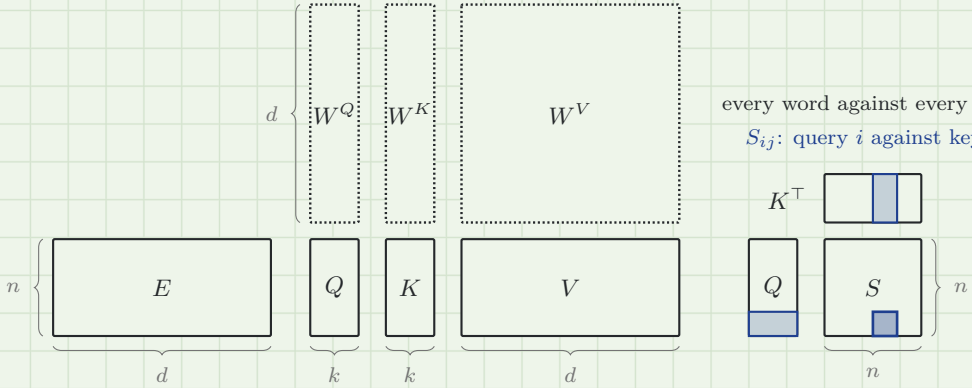
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In a trained transformer nobody writes the weights: they are learned from text, and every head of every layer draws its own map of which words read which.

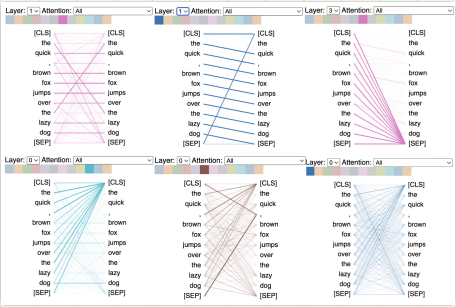


Image from https://www.comet.com/site/blog/explainable-ai-for-transformers/

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Ties such as *bootet kat dukk* are broken by the position of every word, added to its row before Q, K and V are made.